

Foundation (Jalapeno)

- Q1.** What is the GCF of 12 and 18?
(A) 2
(B) 3
(C) 6
(D) 9
(E) 12
- Q2.** A space lab has 15 packets of oxygen and 20 packets of food. What is the LCM of 15 and 20?
(A) 30
(B) 60
(C) 75
(D) 100
(E) 120
- Q3.** What is the GCF of 24 and 30 using prime factorization?
(A) 2
(B) 3
(C) 6
(D) 12
(E) 24
- Q4.** A physics experiment requires 8 repetitions of a task. If each repetition takes 12 minutes, what is the LCM of 8 and 12?
(A) 12
(B) 16
(C) 24
(D) 32
(E) 48
- Q5.** What is the GCF of 9 and 15?
(A) 1
(B) 3
(C) 5
(D) 9
(E) 15
- Q6.** A rocket has 18 communication satellites and 24 navigation satellites. What is the GCF of 18 and 24?
(A) 2
(B) 3
(C) 6
(D) 9
(E) 12
- Q7.** What is the LCM of 9 and 12 using multiples?
(A) 18
(B) 24
(C) 27
(D) 36
(E) 48
- Q8.** A space station has 20 modules with 15 astronauts in each module. What is the LCM of 20 and 15?
(A) 30
(B) 45
(C) 60
(D) 75
(E) 100

Open Ended Questions (FRQ)

- Q9.** A lab experiment requires 12 containers of a chemical and 18 containers of another chemical. If each container of the first chemical weighs 3 kg and each container of the second chemical weighs 2 kg, what is the total weight of the chemicals if the GCF of 12 and 18 is used to optimize storage?
- Q10.** A space mission has 15 crew members who need to be divided into teams of 3 or 5 for training. If the LCM of 3 and 5 is used to determine the number of teams, how many teams can be formed and what is the total number of crew members in these teams?

Chef's Tips

- Ensure students use prime factorization or multiples to find GCF and LCM
- Encourage students to read the questions carefully and choose the correct units
- Remind students to show their work for the free-response questions